VISVESVARAYA TECHNOLOGICAL UNIVERSITY

"JnanaSangama", Belgaum- 590014, Karnataka.

LAB RECORD

on

Big Data Analytics (23CS6PCBDA)

Submitted by

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in partial fulfillment for the award of the degree of

BACHELOR OF ENGINEERING

in

COMPUTER SCIENCE AND ENGINEERING

B.M.S. COLLEGE OF ENGINEERING
(Autonomous Institution under VTU)
BENGALURU - 560019
January 2026 – May 2026

B.M.S. College of Engineering

Bull Temple Road, Bangalore 560019
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Department of Computer Science and Engineering

CERTIFICATE

This is to certify that the Lab work entitled "Big Data Analytics" carried out by **Vinayak Sunil Rodd (1WA23CS045),** who is bonafide student of **B.M.S. College of Engineering.** It is in partial fulfilment for the award of **Bachelor of Engineering in Computer Science and Engineering** of the Visvesvaraya Technological University, Belgaum during the year 2026. The Lab report has been approved as it satisfies the academic requirements in respect of a Big Data Analytics (23CS6PCBDA) work prescribed for the said degree.

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		• Create a column family by name
		  o Employee-Info with attributes
		  o Emp_Id Primary Key, Emp_Name, Designation,
		  o Date_of_Joining, Salary, Dept_Name
		• Insert the values into the table in batch
		• Update Employee name and Department of EmpId 121
		• Sort the details of Employee records based on salary
		• Alter the schema of the table Employee_Info to add a column Projects which stores a set of Projects done by the corresponding Employee.
		• Update the altered table to add project names.
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3	13.02.26	Perform the following DB operations using Cassandra.
		• Create a keyspace by name Library
		• Create a column family by name Library-Info with attributes
		  o Stud_Id Primary Key,
		  o Counter_value of type Counter,
		  o Stud_Name, Book-Name, Book-Id,
		  o Date_of_issue
		• Insert the values into the table in batch
		• Display the details of the table created and increase the value of the counter
		• Write a query to show that a student with id 112 has taken a book “BDA” 2 times.
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Github Link: https://github.com/Vinayak1205/BDALab/tree/main

Course Outcomes (COs):

CO1	Apply the concept of NoSQL, Hadoop or Spark for a given task
CO2	Analyse big data analytics mechanisms that can be applied to obtain solution for a given problem.
CO3	Design and implement solutions using data analytics mechanisms for a given problem.

LABORATORY PROGRAM – 1

MongoDB- CRUD Operations Demonstration (Practice and Self Study)

COMMAND WITH OUTPUT - USING ATLAS

```
test> use BDA_LAB
switched to db BDA_LAB
BDA_LAB> db.createCollection("Employee")
{ ok: 1 }
BDA_LAB> db.Employee.insertOne({
...     Emp_Id: 101,
...     Name: "Manoj",
...     Dept: "CSE",
...     Salary: 50000
... })
{
  acknowledged: true,
  insertedId: ObjectId('69f5bb2311d23b4927149f48')
}
BDA_LAB> db.Employee.insertMany([
...     {Emp_Id: 102, Name: "Rahul", Dept: "ISE", Salary: 40000},
...     {Emp_Id: 103, Name: "Anjali", Dept: "ECE", Salary: 45000},
...     {Emp_Id: 104, Name: "Kiran", Dept: "CSE", Salary: 55000}
... ])
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('69f5bb2a11d23b4927149f49'),
    '1': ObjectId('69f5bb2a11d23b4927149f4a'),
    '2': ObjectId('69f5bb2a11d23b4927149f4b')
  }
}
BDA_LAB> db.Employee.find()
[
  {
    _id: ObjectId('69f5bb2311d23b4927149f48'),
    Emp_Id: 101,
    Name: 'Manoj',
```

```
  }
}
BDA_LAB> db.Employee.find()
[
  {
    _id: ObjectId('69f5bb2311d23b4927149f48'),
    Emp_Id: 101,
    Name: 'Manoj',
    Dept: 'CSE',
    Salary: 50000
  },
  {
    _id: ObjectId('69f5bb2a11d23b4927149f49'),
    Emp_Id: 102,
    Name: 'Rahul',
    Dept: 'ISE',
    Salary: 40000
  },
  {
    _id: ObjectId('69f5bb2a11d23b4927149f4a'),
    Emp_Id: 103,
    Name: 'Anjali',
    Dept: 'ECE',
    Salary: 45000
  },
  {
    _id: ObjectId('69f5bb2a11d23b4927149f4b'),
    Emp_Id: 104,
    Name: 'Kiran',
    Dept: 'CSE',
    Salary: 55000
  }
]
BDA_LAB>
```

```
BDA_LAB> db.Employee.updateOne(
...      {Emp_Id: 101},
...      {$set: {Salary: 60000}}
... )
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
BDA_LAB> db.Employee.updateMany(
...      {Dept: "CSE"},
...      {$set: {Dept: "AI"}}
... )
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 2,
  modifiedCount: 2,
  upsertedCount: 0
}
BDA_LAB>
```

LABORATORY PROGRAM – 2

Perform the following DB operations using Cassandra

Questions:

a) Create a keyspace by name Employee

b) Create a column family by name

- Employee-Info with attributes
- Emp_Id Primary Key, Emp_Name,
- Designation, Date_of_Joining,
- Salary, Dept_Name

c) Insert the values into the table in batch

d) Update Employee name and Department of Emp-Id 121

e) Sort the details of Employee records based on salary

f) Alter the schema of the table Employee_Info to add a column Projects which stores a set of Projects done by the corresponding Employee.

g) Update the altered table to add project names.

h) Create a TTL of 15 seconds to display the values of Employees.

COMMAND WITH OUTPUT

```
cqlsh> CREATE KEYSPACE Employee
   ... WITH replication = {'class': 'SimpleStrategy', 'replication_factor': 1};

cqlsh>
cqlsh> USE Employee;
cqlsh:employee> CREATE TABLE Employee_Info (
            ...     Emp_Id INT PRIMARY KEY,
            ...     Emp_Name TEXT,
            ...     Designation TEXT,
            ...     Date_of_Joining DATE,
            ...     Salary INT,
            ...     Dept_Name TEXT
            ... );
cqlsh:employee> BEGIN BATCH
            ... INSERT INTO Employee_Info (Emp_Id, Emp_Name, Designation, Date_of_Joining, Salary, Dept_Name)
            ... VALUES (121, 'Manoj', 'Engineer', '2022-01-10', 50000, 'CSE');
            ...
            ... INSERT INTO Employee_Info (Emp_Id, Emp_Name, Designation, Date_of_Joining, Salary, Dept_Name)
            ... VALUES (122, 'Rahul', 'Manager', '2021-05-15', 70000, 'ISE');
            ...
            ... INSERT INTO Employee_Info (Emp_Id, Emp_Name, Designation, Date_of_Joining, Salary, Dept_Name)
            ... VALUES (123, 'Anjali', 'Analyst', '2023-03-20', 45000, 'ECE');
            ... APPLY BATCH;
cqlsh:employee> UPDATE Employee_Info
            ... SET Emp_Name = 'Manoj Kumar', Dept_Name = 'AI'
            ... WHERE Emp_Id = 121;
cqlsh:employee> SELECT * FROM Employee_Info ORDER BY Salary DESC ALLOW FILTERING;
```

```
cqlsh:employee> ALTER TABLE Employee_Info
            ... ADD Projects SET<TEXT>;
cqlsh:employee> UPDATE Employee_Info
            ... SET Projects = {'Hadoop', 'Spark'}
            ... WHERE Emp_Id = 121;
cqlsh:employee>
cqlsh:employee> UPDATE Employee_Info
            ... SET Projects = {'ML', 'Data Mining'}
            ... WHERE Emp_Id = 122;
cqlsh:employee>
```

```
cqlsh:employee> INSERT INTO Employee_Info (Emp_Id, Emp_Name, Designation, Date_of_Joining, Salary, Dept_Name)
            ... VALUES (130, 'TempUser', 'Intern', '2024-01-01', 20000, 'CSE')
            ... USING TTL 15;
cqlsh:employee> SELECT * FROM Employee_Info;

 emp_id | date_of_joining | dept_name | designation | emp_name    | projects                | salary
--------+-----------------+-----------+-------------+-------------+-------------------------+--------
    123 |      2023-03-20 |       ECE |     Analyst |      Anjali |                    null |  45000
    130 |      2024-01-01 |       CSE |      Intern |    TempUser |                    null |  20000
    122 |      2021-05-15 |       ISE |     Manager |       Rahul | {'Data Mining', 'ML'} |  70000
    121 |      2022-01-10 |        AI |    Engineer | Manoj Kumar |   {'Hadoop', 'Spark'} |  50000

(4 rows)
cqlsh:employee> SELECT * FROM Employee_Info;

 emp_id | date_of_joining | dept_name | designation | emp_name    | projects                | salary
--------+-----------------+-----------+-------------+-------------+-------------------------+--------
    123 |      2023-03-20 |       ECE |     Analyst |      Anjali |                    null |  45000
    130 |      2024-01-01 |       CSE |      Intern |    TempUser |                    null |  20000
    122 |      2021-05-15 |       ISE |     Manager |       Rahul | {'Data Mining', 'ML'} |  70000
    121 |      2022-01-10 |        AI |    Engineer | Manoj Kumar |   {'Hadoop', 'Spark'} |  50000

(4 rows)
```

LABORATORY PROGRAM – 3

Perform the following DB operations using Cassandra

Questions:

a) Create a keyspace by name Library
b) Create a column family by name Library-Info with attributes

- Stud_Id Primary Key,
- Counter_value of type Counter,
- Stud_Name, Book-Name, Book-Id,
- Date_of_issue

c) Insert the values into the table in batch
d) Display the details of the table created and increase the value of the counter
e) Write a query to show that a student with id 112 has taken a book "BDA" 2 times.
f) Export the created column to a csv file
g) Import a given csv dataset from local file system into Cassandra column family

COMMAND WITH OUTPUT

```
cqlsh> CREATE KEYSPACE Library
   ... WITH replication = {'class': 'SimpleStrategy', 'replication_factor': 1};
cqlsh> USE Library;
cqlsh:library> CREATE TABLE Library_Info (
           ...     Stud_Id INT PRIMARY KEY,
           ...     Stud_Name TEXT,
           ...     Book_Name TEXT,
           ...     Book_Id INT,
           ...     Date_of_issue DATE
           ... );
cqlsh:library> CREATE TABLE Book_Counter (
           ...     Stud_Id INT,
           ...     Book_Name TEXT,
           ...     Counter_Value COUNTER,
           ...     PRIMARY KEY (Stud_Id, Book_Name)
           ... );
cqlsh:library> BEGIN BATCH
           ... INSERT INTO Library_Info (Stud_Id, Stud_Name, Book_Name, Book_Id, Date_of_issue)
           ... VALUES (111, 'Manoj', 'BDA', 201, '2024-04-01');
           ...
           ... INSERT INTO Library_Info (Stud_Id, Stud_Name, Book_Name, Book_Id, Date_of_issue)
           ... VALUES (112, 'Rahul', 'BDA', 202, '2024-04-02');
           ...
           ... INSERT INTO Library_Info (Stud_Id, Stud_Name, Book_Name, Book_Id, Date_of_issue)
           ... VALUES (113, 'Anjali', 'DBMS', 203, '2024-04-03');
           ... APPLY BATCH;
cqlsh:library> UPDATE Book_Counter
           ... SET Counter_Value = Counter_Value + 1
           ... WHERE Stud_Id = 112 AND Book_Name = 'BDA';
cqlsh:library>
cqlsh:library> UPDATE Book_Counter
           ... SET Counter_Value = Counter_Value + 1
           ... WHERE Stud_Id = 112 AND Book_Name = 'BDA';
cqlsh:library>
```

```
cqlsh:library> SELECT * FROM Library_Info;

 stud_id | book_id | book_name | date_of_issue | stud_name
---------+---------+-----------+---------------+----------
     111 |     201 |       BDA |    2024-04-01 |     Manoj
     113 |     203 |      DBMS |    2024-04-03 |    Anjali
     112 |     202 |       BDA |    2024-04-02 |     Rahul

(3 rows)
cqlsh:library> SELECT * FROM Book_Counter;

 stud_id | book_name | counter_value
---------+-----------+---------------
     112 |       BDA |             2

(1 rows)
cqlsh:library> SELECT * FROM Book_Counter
           ... WHERE Stud_Id = 112 AND Book_Name = 'BDA';

 stud_id | book_name | counter_value
---------+-----------+---------------
     112 |       BDA |             2

(1 rows)
cqlsh:library> cqlsh -e "COPY Library.Library_Info TO 'library.csv' WITH HEADER = TRUE;"
           ...
cqlsh:library>
```

LABORATORY PROGRAM – 4

Execution of HDFS Commands for interaction with Hadoop Environment. (Minimum 10 commands to be executed)

COMMAND WITH OUTPUT

```
Starting namenodes on [localhost]
Starting datanodes
```

```
Starting resourcemanager
Starting nodemanagers
```

```
~/Downloads$ hdfs dfs -mkdir /input_file
~/Downloads$ echo "Hello world!" > input.txt
~/Downloads$ hdfs dfs -put input.txt /input_file
~/Downloads$ hdfs dfs -ls /input_file

group          13 2026-05-05 00:29 /input_file/input.txt
~/Downloads$ hdfs dfs -cat /input_file/input.txt

~/Downloads$ hdfs dfs -rm -r /input_file/input.txt

~/Downloads$ hdfs dfs -ls /input_file
~/Downloads$ echo "Hello\nWorld" > input1.txt
~/Downloads$ hdfs dfs -cat /input1.txt
ile or directory
~/Downloads$ hdfs dfs -put input1.txt /input_file
~/Downloads$ hdfs dfs -cat /input_file/input1.txt

~/Downloads$ hdfs dfs -rm -r /input_file

~/Downloads$
```

LABORATORY PROGRAM – 5
Implement Wordcount program on Hadoop framework

```
import java.io.Exception
importorg.apache.hadoop.conf.Configured
; import org.apache.hadoop.fs.Path;
importorg.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;
importorg.apache.hadoop.mapred.FileInputFormat;
importorg.apache.hadoop.mapred.FileOutputFormat;
import org.apache.hadoop.mapred.JobClient;
importorg.apache.hadoop.mapred.JobConf;
import org.apache.hadoop.util.Tool;
import org.apache.hadoop.util.ToolRunner;

public class WCDriver extends Configured implements Tool

  { public int run(String[] args) throws IOException {

    if (args.length < 2) {
      System.out.println("Please give valid inputs");
      return -1;
    }

    JobConf conf = new JobConf(WCDriver.class);
    conf.setJobName("WordCount");

    FileInputFormat.setInputPaths(conf, new Path(args[0]));
    FileOutputFormat.setOutputPath(conf, new
    Path(args[1]));

    conf.setMapperClass(WCMapper.clas
    s);
    conf.setReducerClass(WCReducer.cla
    ss);

    conf.setMapOutputKeyClass(Text.class);
    conf.setMapOutputValueClass(IntWritable.class
    );

    conf.setOutputKeyClass(Text.class);
    conf.setOutputValueClass(IntWritable.class);

    JobClient.runJob(conf);
```

```
    return 0;
}

// Main Method
```

```
    public static void main(String[] args) throws
      Exception { int exitCode = ToolRunner.run(new
      WCDriver(), args); System.out.println("Job Exit
      Code: " + exitCode);
    }
  }}
```

Mapper Code:

```
// Importing libraries
import
java.io.IOException;

import org.apache.hadoop.io.IntWritable;
import
org.apache.hadoop.io.LongWritable;
import org.apache.hadoop.io.Text;
import
org.apache.hadoop.mapred.MapReduceBase;
import org.apache.hadoop.mapred.Mapper;
import
org.apache.hadoop.mapred.OutputCollector;
import org.apache.hadoop.mapred.Reporter;

public class WCMapper extends MapReduceBase
    implements Mapper<LongWritable, Text, Text, IntWritable> {

  // Map function
  public void map(LongWritable key, Text value,
            OutputCollector<Text, IntWritable>
            output, Reporter reporter) throws
            IOException {

    String line = value.toString();

    // Splitting the line on whitespace
    for (String word : line.split("\\s+"))
    {
      if (word.length() > 0) {
        output.collect(new Text(word), new IntWritable(1));
      }
    }
  }
}
```

Reducer Code

```
// Importing libraries
import
java.io.IOException;
import java.util.Iterator;
```

```
import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;
import
org.apache.hadoop.mapred.MapReduceBase;
import  org.apache.hadoop.mapred.OutputCollector;
import org.apache.hadoop.mapred.Reducer;
import org.apache.hadoop.mapred.Reporter;
```

```
public class WCReducer extends MapReduceBase
        implements Reducer<Text, IntWritable, Text, IntWritable> {

    // Reduce function
    public void reduce(Text key, Iterator<IntWritable> values,
                       OutputCollector<Text, IntWritable> output,
                       Reporter reporter) throws IOException {

        int count = 0;

        // Counting the frequency of each word
        while (values.hasNext()) {
            count += values.next().get();
        }

        output.collect(key, new IntWritable(count));
    }
}
```

```
2026-05-05 00:41:05,096 INFO client.DefaultNoHARMFailoverProxyProvider: Connecting to ResourceManager at /0.0.0.0:8032
2026-05-05 00:41:06,216 WARN mapreduce.JobResourceUploader: Hadoop command-line option parsing not performed. Implement the Tool interface and execute your application with ToolRunner to remedy this.
2026-05-05 00:41:06,257 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/yatharth/.staging/job_1777921102769_0001
2026-05-05 00:41:06,726 INFO input.FileInputFormat: Total input files to process : 1
2026-05-05 00:41:07,272 INFO mapreduce.JobSubmitter: number of splits:1
2026-05-05 00:41:07,700 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1777921102769_0001
2026-05-05 00:41:07,700 INFO mapreduce.JobSubmitter: Executing with tokens: []
2026-05-05 00:41:08,026 INFO conf.Configuration: resource-types.xml not found
2026-05-05 00:41:08,031 INFO resource.ResourceUtils: Unable to find 'resource-types.xml'.
2026-05-05 00:41:10,366 INFO impl.YarnClientImpl: Submitted application application_1777921102769_0001
2026-05-05 00:41:10,557 INFO mapreduce.Job: The url to track the job: http://yatharth-VirtualBox:8088/proxy/application_1777921102769_0001/
2026-05-05 00:41:10,566 INFO mapreduce.Job: Running job: job_1777921102769_0001
2026-05-05 00:41:28,454 INFO mapreduce.Job: Job job_1777921102769_0001 running in uber mode : false
2026-05-05 00:41:28,466 INFO mapreduce.Job:  map 0% reduce 0%
2026-05-05 00:41:42,061 INFO mapreduce.Job:  map 100% reduce 0%
2026-05-05 00:41:48,139 INFO mapreduce.Job:  map 100% reduce 100%
2026-05-05 00:41:49,164 INFO mapreduce.Job: Job job_1777921102769_0001 completed successfully
2026-05-05 00:41:49,317 INFO mapreduce.Job: Counters: 54
        File System Counters
                FILE: Number of bytes read=126
                FILE: Number of bytes written=551777
                FILE: Number of read operations=0
                FILE: Number of large read operations=0
                FILE: Number of write operations=0
                HDFS: Number of bytes read=163
                HDFS: Number of bytes written=76
```

```
=       1
analysis.       1
bda     1
bda.    1
big     1
data    1
hadoop  1
hello   2
in      1
is      1
this    1
```

LABORATORY PROGRAM – 6

Implement Weather program on Hadoop framework

Questions:

From the following link extract the weather data
https://github.com/tomwhite/hadoopbook/tree/master/input/ncdc/all

a) Create a MapReduce program to find average temperature for each year from NCDC data set.

b) find the mean max temperature for every month.

CODE, COMMAND WITH OUTPUT – A
Driver Code

```
package temp;

import
org.apache.hadoop.conf.Configuration;
import org.apache.hadoop.fs.Path;
import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;
import
org.apache.hadoop.mapreduce.Job;
import org.apache.hadoop.mapreduce.lib.input.FileInputFormat;
import org.apache.hadoop.mapreduce.lib.output.FileOutputFormat;

public class AverageDriver {
   public static void main(String[] args) throws Exception {
      if (args.length != 2) {
         System.err.println("Please enter both input and output
         parameters."); System.exit(-1);
      }

      // Creating a configuration and job instance
      Configuration conf = new Configuration();
      Job job = Job.getInstance(conf, "Average Calculation");
      job.setJarByClass(AverageDriver.class);

      // Input and output paths
      FileInputFormat.addInputPath(job, new Path(args[0]));
      FileOutputFormat.setOutputPath(job, new
      Path(args[1]));

      // Setting mapper and reducer classes
```

```
job.setMapperClass(AverageMapper.class);
job.setReducerClass(AverageReducer.class)
;
```

```
        // Output key and value types
        job.setOutputKeyClass(Text.class);
        job.setOutputValueClass(IntWritable.class)
        ;

        // Submitting the job and waiting for it to complete
        System.exit(job.waitForCompletion(true) ? 0 : 1);
    }
}
```

MAPPER CODE

```
package temp;


import java.io.IOException;


import org.apache.hadoop.io.IntWritable;
import
org.apache.hadoop.io.LongWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Mapper;


public class AverageMapper extends Mapper<LongWritable, Text, Text, IntWritable> {


    public static final int MISSING = 9999;


    @Override
    public void map(LongWritable key, Text value, Context context)
            throws IOException, InterruptedException {


        String line = value.toString();


        // Extract year from fixed position
        String year = line.substring(15, 19);
        int temperature;
```

```
// Determine if there's a '+' sign
if (line.charAt(87) == '+') {
```

```
            temperature = Integer.parseInt(line.substring(88, 92));
        } else {
            temperature = Integer.parseInt(line.substring(87, 92));
        }

        // Quality check character
        String quality = line.substring(92, 93);

        // Only emit if data is valid
        if (temperature != MISSING && quality.matches("[01459]")) {
            context.write(new Text(year), new IntWritable(temperature));
        }
    }
```

REDUCER CODE

```
package temp;

import java.io.IOException;

import
org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Reducer;

public class AverageReducer extends Reducer<Text, IntWritable, Text, IntWritable> {

    @Override
    public void reduce(Text key, Iterable<IntWritable> values, Context context)
            throws IOException, InterruptedException {

        int sumTemp = 0;
        int count = 0;
```

```
        for (IntWritable value : values) {
            sumTemp += value.get();
            count++;
        }

        if (count > 0) {
            int average = sumTemp / count;
            context.write(key, new IntWritable(average));
        }
    }
```

```
2026-05-05 01:09:16,335 INFO client.DefaultNoHARMFailoverProxyProvider: Connecting to ResourceManager at /0.0.0.0:8032
2026-05-05 01:09:17,266 WARN mapreduce.JobResourceUploader: Hadoop command-line option parsing not performed. Implement the Tool interf
2026-05-05 01:09:17,346 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/yatharth/.stagi
2026-05-05 01:09:17,825 INFO input.FileInputFormat: Total input files to process : 1
2026-05-05 01:09:18,452 INFO mapreduce.JobSubmitter: number of splits:1
2026-05-05 01:09:18,799 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1777921102769_0002
2026-05-05 01:09:18,800 INFO mapreduce.JobSubmitter: Executing with tokens: []
2026-05-05 01:09:19,036 INFO conf.Configuration: resource-types.xml not found
2026-05-05 01:09:19,037 INFO resource.ResourceUtils: Unable to find 'resource-types.xml'.
2026-05-05 01:09:19,955 INFO impl.YarnClientImpl: Submitted application application_1777921102769_0002
2026-05-05 01:09:20,038 INFO mapreduce.Job: The url to track the job: http://yatharth-VirtualBox:8088/proxy/application_1777921102769_0
2026-05-05 01:09:20,039 INFO mapreduce.Job: Running job: job_1777921102769_0002
2026-05-05 01:09:31,711 INFO mapreduce.Job: Job job_1777921102769_0002 running in uber mode : false
2026-05-05 01:09:31,712 INFO mapreduce.Job:  map 0% reduce 0%
2026-05-05 01:09:37,911 INFO mapreduce.Job:  map 100% reduce 0%
2026-05-05 01:09:43,988 INFO mapreduce.Job:  map 100% reduce 100%
2026-05-05 01:09:44,021 INFO mapreduce.Job: Job job_1777921102769_0002 completed successfully
2026-05-05 01:09:44,147 INFO mapreduce.Job: Counters: 54
        File System Counters
                FILE: Number of bytes read=71
                FILE: Number of bytes written=551351
                FILE: Number of read operations=0
                FILE: Number of large read operations=0
                FILE: Number of write operations=0
                HDFS: Number of bytes read=145
                HDFS: Number of bytes written=24
                HDFS: Number of read operations=8
                HDFS: Number of large read operations=0
                HDFS: Number of write operations=2
                HDFS: Number of bytes read erasure-coded=0
```

LABORATORY PROGRAM – 7

For a given Text file, Create a Map Reduce program to sort the content in an alphabetic order listing only top 10 maximum occurrences of words.

```
import
org.apache.hadoop.conf.Configuration;
import org.apache.hadoop.fs.Path;

import
org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;

import org.apache.hadoop.mapreduce.Job;
import org.apache.hadoop.mapreduce.lib.input.FileInputFormat;
import org.apache.hadoop.mapreduce.lib.output.FileOutputFormat;

public class TopNDriver {
   public static void main(String[] args) throws Exception {
      if (args.length != 3) {
         System.err.println("Usage: TopNDriver <in> <temp-out> <final-out>");
         System.exit(2);
      }

      Configuration conf = new Configuration();

      // === Job 1: Word Count ===
      Job wcJob = Job.getInstance(conf, "word count");
      wcJob.setJarByClass(TopNDriver.class);
      wcJob.setMapperClass(WordCountMapper.class);
      wcJob.setCombinerClass(WordCountReducer.class)
      ;
      wcJob.setReducerClass(WordCountReducer.class);
      wcJob.setOutputKeyClass(Text.class);
      wcJob.setOutputValueClass(IntWritable.class);

      FileInputFormat.addInputPath(wcJob, new Path(args[0]));
      Path tempDir = new Path(args[1]);
      FileOutputFormat.setOutputPath(wcJob, tempDir);

      if (!wcJob.waitForCompletion(true))
         { System.exit(1);
      }
```

// === Job 2: Top N ===

```
        Job topJob = Job.getInstance(conf, "top 10 words");
        topJob.setJarByClass(TopNDriver.class);
        topJob.setMapperClass(TopNMapper.class);
        topJob.setReducerClass(TopNReducer.class);
        topJob.setMapOutputKeyClass(IntWritable.class);
        topJob.setMapOutputValueClass(Text.class);
        topJob.setOutputKeyClass(Text.class);
        topJob.setOutputValueClass(IntWritable.class);

        FileInputFormat.addInputPath(topJob, tempDir);
        FileOutputFormat.setOutputPath(topJob, new
        Path(args[2]));

        System.exit(topJob.waitForCompletion(true) ? 0 : 1);
    }
}
```

Mapper Code (WordCountMapper.java)

```
package samples.topn;

import java.io.IOException;
import
java.util.StringTokenizer;

import
org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Mapper;
public class WordCountMapper extends Mapper<Object, Text, Text, IntWritable> {
    private static final IntWritable ONE = new IntWritable(1);
    private Text word = new Text();

    // characters to normalize into spaces
    private String tokens = "[_|$#<>\\^=\\[\\]\\*/\\\\,;,.\\-:()?!\"']";

    @Override
    protected void map(Object key, Text value, Context context)
            throws IOException, InterruptedException {

        // clean & tokenize
        String clean = value.toString()
                .toLowerCase()
                .replaceAll(tokens, " ");
```

```
StringTokenizer itr = new StringTokenizer(clean);
```

```
        while (itr.hasMoreTokens()) {
            word.set(itr.nextToken().trim())
            ; context.write(word, ONE);
        }
    }
}
```

Mapper Code (TopNMapper.java)

```
package samples.topn;

import

java.io.IOException;

import
org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Mapper;
public class TopNMapper extends Mapper<Object, Text, IntWritable, Text> {
    private IntWritable count = new IntWritable();
    private Text word = new Text();

    @Override
    protected void map(Object key, Text value, Context context)
            throws IOException, InterruptedException {

        // input line: word \t count
        String[] parts = value.toString().split("\\t");

        if (parts.length == 2) {
            word.set(parts[0]);
            count.set(Integer.parseInt(parts[1])
            );

            // emit count → word, so Hadoop sorts by count
            context.write(count, word);
        }
    }
}
```

Reducer Code (WordCountReducer.java)

```
package samples.topn;

import
```

```
java.io.IOException;

import
org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Reducer;
```

```
public class WordCountReducer extends Reducer<Text, IntWritable, Text, IntWritable> {

    @Override
    protected void reduce(Text key, Iterable<IntWritable> values, Context context)
            throws IOException, InterruptedException {

        int sum = 0;

        for (IntWritable val : values) {
            sum += val.get();
        }

        context.write(key, new IntWritable(sum));
    }
}
```

Reducer Code (TopNReducer.java)

```
package samples.topn;

import
java.io.IOException;
import java.util.ArrayList;
import
java.util.Collections;
import java.util.List;
import java.util.Map;
import
java.util.TreeMap;

import
org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Reducer;

public class TopNReducer extends Reducer<IntWritable, Text, Text, IntWritable> {

    // TreeMap with descending order of keys (counts)
    private TreeMap<Integer, List<String>> countMap =
            new TreeMap<>(Collections.reverseOrder());

    @Override
    protected void reduce(IntWritable key, Iterable<Text> values, Context context)
            throws IOException, InterruptedException {
```

```
int cnt = key.get();

List<String> words = countMap.getOrDefault(cnt, new ArrayList<>());
```

```
        for (Text w : values) {
            words.add(w.toString())
            ;
        }

        countMap.put(cnt, words);
    }

    @Override
    protected void cleanup(Context context)
            throws IOException, InterruptedException {

        // collect top 10 word→count pairs
        List<WordCount> topList = new ArrayList<>();
        int seen = 0;

        for (Map.Entry<Integer, List<String>> entry : countMap.entrySet()) {
            int cnt = entry.getKey();

            for (String w : entry.getValue()) {
                topList.add(new WordCount(w,
                cnt)); seen++;

                if (seen == 10) break;
            }

            if (seen == 10) break;
        }

        // emit results
        for (WordCount wc : topList) {
            context.write(new Text(wc.getWord()), new IntWritable(wc.getCount()));
        }
    }
}
        // sort these 10 entries alphabetically by word
        Collections.sort(topList, (a, b) -> a.word.compareTo(b.word));

        // emit final top 10 in alphabetical order
        for (WordCount wc : topList) {
            context.write(new Text(wc.word), new IntWritable(wc.count));
        }
    }
```

// helper class

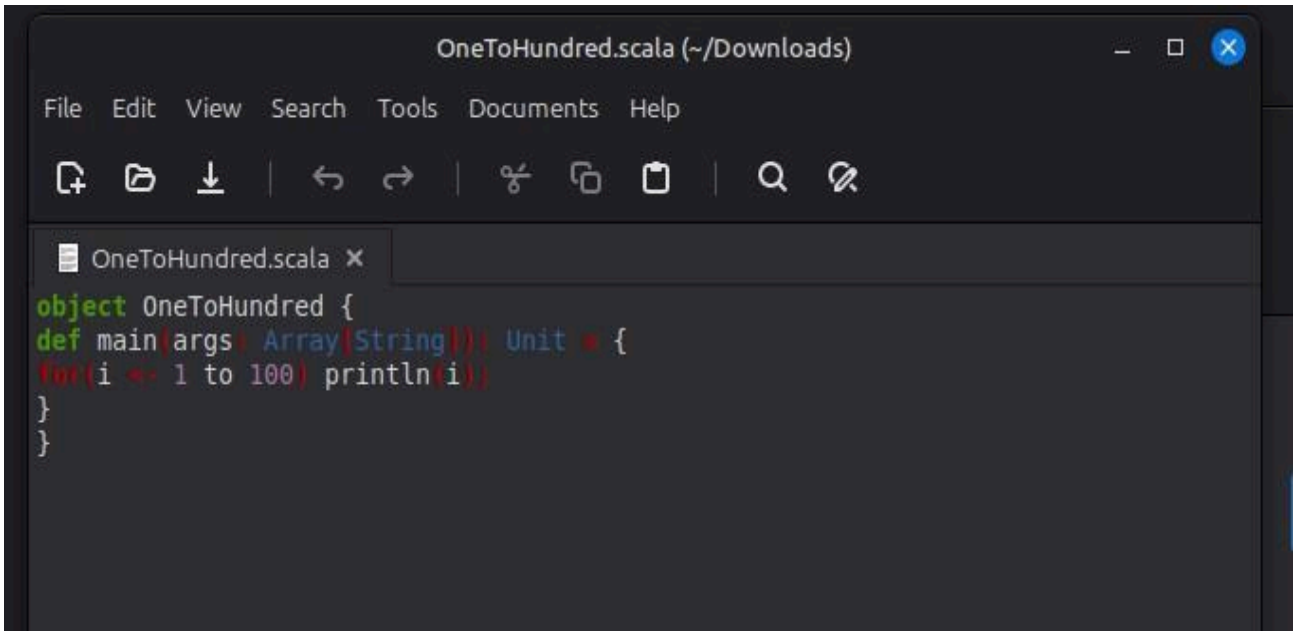
```
    private static class WordCount {
        String word;
        int count;

        WordCount(String w, int c) {
            word = w;
            count = c;
        }
    }
}
```

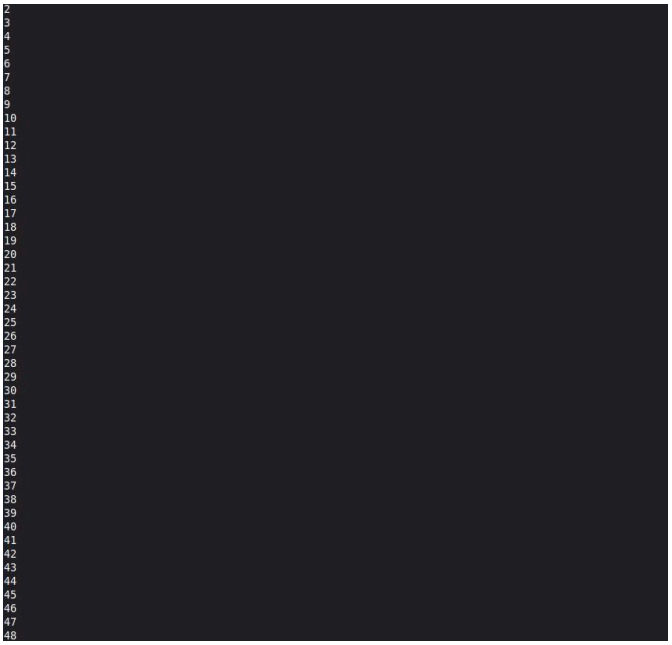
```
2026-05-05 01:26:39,747 INFO client.DefaultNoHARMFailoverProxyProvider: Connecting to ResourceManager at /0.0.0.0:8032
2026-05-05 01:26:40,113 WARN mapreduce.JobResourceUploader: Hadoop command-line option parsing not performed. Implement the Tool interface and execute you
2026-05-05 01:26:40,150 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/yatharth/.staging/job_1777921102769
2026-05-05 01:26:40,491 INFO input.FileInputFormat: Total input files to process : 1
2026-05-05 01:26:40,621 INFO mapreduce.JobSubmitter: number of splits:1
2026-05-05 01:26:40,822 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1777921102769_0003
2026-05-05 01:26:40,822 INFO mapreduce.JobSubmitter: Executing with tokens: []
2026-05-05 01:26:40,993 INFO conf.Configuration: resource-types.xml not found
2026-05-05 01:26:40,994 INFO resource.ResourceUtils: Unable to find 'resource-types.xml'.
2026-05-05 01:26:41,110 INFO impl.YarnClientImpl: Submitted application application_1777921102769_0003
2026-05-05 01:26:41,189 INFO mapreduce.Job: The url to track the job: http://yatharth-VirtualBox:8088/proxy/application_1777921102769_0003/
2026-05-05 01:26:41,190 INFO mapreduce.Job: Running job: job_1777921102769_0003
2026-05-05 01:26:49,456 INFO mapreduce.Job: Job job_1777921102769_0003 running in uber mode : false
2026-05-05 01:26:49,458 INFO mapreduce.Job:  map 0% reduce 0%
2026-05-05 01:26:54,562 INFO mapreduce.Job:  map 100% reduce 0%
2026-05-05 01:27:01,679 INFO mapreduce.Job:  map 100% reduce 100%
2026-05-05 01:27:01,697 INFO mapreduce.Job: Job job_1777921102769_0003 completed successfully
2026-05-05 01:27:01,850 INFO mapreduce.Job: Counters: 54
        File System Counters
                FILE: Number of bytes read=325
                FILE: Number of bytes written=551835
                FILE: Number of read operations=0
                FILE: Number of large read operations=0
                FILE: Number of write operations=0
                HDFS: Number of bytes read=260
                HDFS: Number of bytes written=41
                HDFS: Number of read operations=8
                HDFS: Number of large read operations=0
                HDFS: Number of write operations=2
```

LABORATORY PROGRAM – 8

Write a Scala program to print numbers from 1 to 100 using for loop.

CODE, COMMAND WITH OUTPUT





LABORATORY PROGRAM – 9

Using RDD and FlatMap count how many times each word appears in a file and write out a list of words whose count is strictly greater than 4 using Spark.

CODE, COMMAND WITH OUTPUT

```
>>> rdd = spark.sparkContext.textFile("input.txt")
>>> words = rdd.flatMap(lambda line: line.split(" "))
>>> pairs = words.map(lambda word: (word, 1))
>>> counts = pairs.reduceByKey(lambda a, b: a + b)
>>> result = counts.filter(lambda x: x[1] > 4)
>>> result.collect()
[('hello', 5), ('spark', 6)]
```

LABORATORY PROGRAM – 10

Write a simple streaming program in Spark to receive text data streams on a particular port, perform basic text cleaning (like white space removal, stop words removal, lemmatization, etc.), and print the cleaned text on the screen. (Open Ended Question).

CODE, COMMAND WITH OUTPUT

```
# Install NLTK and download required data (run once)
!pip install nltk

import nltk
nltk.download('punkt')
nltk.download('stopwords'
)
nltk.download('wordnet')

from pyspark.sql import SparkSession
from pyspark.sql.functions import col, lower, regexp_replace, split, explode, udf
from pyspark.sql.types import ArrayType, StringType
from pyspark.ml.feature import StopWordsRemover
from nltk.stem import WordNetLemmatizer

# Initialize SparkSession
spark = SparkSession.builder.appName("TextProcessing").getOrCreate()

# Define your input lines
lines = [
    "Hello, I hate you.",
    "I hate that I love you.",
    "Don't want to, but I can't
    put", "nobody else above
    you."
]

# Create DataFrame from lines
df = spark.createDataFrame(lines, "string").toDF("value")

# Step 1: Lowercase and remove punctuation
df_clean = df.select(
    regexp_replace(lower(col("value")), "[^a-zA-Z\\s]", "").alias("cleaned")
```

)

```
# Step 2: Tokenize the cleaned text
df_tokens = df_clean.select(
    split(col("cleaned"), "\\s+").alias("tokens")
)

# Step 3: Remove stop words
remover = StopWordsRemover(inputCol="tokens", outputCol="filtered")
df_filtered = remover.transform(df_tokens)

# Step 4: Lemmatization using NLTK WordNetLemmatizer with UDF
lemmatizer = WordNetLemmatizer()

def lemmatize_words(words):
    return [lemmatizer.lemmatize(word) for word in words]
lemmatize_udf = udf(lemmatize_words, ArrayType(StringType()))
df_lemmatized = df_filtered.withColumn(
    "lemmatized",
    lemmatize_udf(col("filtered")
    )
)

# Step 5: Explode the lemmatized words and show results
df_lemmatized.select(
    explode(col("lemmatized")).alias("word")
).show(truncate=False)
```